

Spreading 34 Points in the Plane

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Abstract

We place 34 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 30.90470+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 35-point configuration gives 32.07769; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 34 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 30.90470\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 34$ is not on the page. Deleting a point from our 35-point configuration gives 32.07769; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n34.txt
```

in <https://github.com/Ge-limin/math719>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
15452354219017296516493805
denominator:
499999999999175627295293
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

34 points, x and y per line. File: friedman-maxmin2d/coords/n34.txt.

```
-6.736633642682 -7.074012235443 -7.206502846312 -1.638987487038
-5.313555263900 -6.068090331541 -8.873165476042 -5.511461881327
-9.879053443371 -5.600695156999 -5.902761276102 -3.319807175734
-7.876441032073 -5.430589248172 -7.733208518099 -7.156707565942
-9.024057795447 -3.792663327237 -6.308309175480 -6.170387229815
-8.261203678961 -1.727948763830 -8.016838833840 -3.704059171728
-10.020782239417 -3.873535960391 -9.194301017938 -2.087572672926
-8.303112212795 -6.334996072307 -9.452382262649 -4.696288332865
-6.450210695175 -4.444898032208 -4.731136994815 -5.255201064890
-7.662958281470 -2.529261714169 -5.771026532627 -5.178865928880
-6.169020661707 -1.822101227331 -9.299836656764 -6.415868705461
-8.444841008840 -4.607836875699 -7.306537337377 -6.252300741808
-4.477054906749 -4.288018407620 -8.596055620447 -2.888885623265
-6.675095499552 -2.684590797413 -7.448116564871 -4.526964242544
-5.474983769822 -4.223691241929 -5.396686438257 -2.457317605653
-7.020114389871 -3.623186538573 -6.878212870175 -5.348675736180
-9.769419036055 -2.905643125416 -4.811821115514 -3.314345029463
```